

Server-side architecture of the auto service CRM web application

Node.js · Express · Prisma · PostgreSQL · JWT · VIN validation · AUTO.RIA API · transactional email

Client tier (React SPA)

HTTP requests, JSON payloads, header Authorization: Bearer <JWT>



Presentation layer

Express middleware

Request parsing (JSON), CORS, logging

Routes

/api/auth, /api/orders, /api/vin, inventory, staff, analytics

REST controllers

Map HTTP verbs to application operations, return JSON

JWT middleware on protected routes before controller logic (“front controller” pattern).



Service & integration layer

Application services

autoRiaService, vinServiceNew, emailService, cacheService outbound HTTPS, caching.

Authentication module

password hashing (bcrypt), issue and validate JWT tokens

Complex workflows reside in services; simple CRUD may call Prisma directly from controllers.



Persistence & external interfaces

Prisma ORM

→ PostgreSQL relational store for customers, vehicles, orders, inventory, purchases, analytics aggregates.

HTTP client (Axios)

AUTO.RIA Developers API; NHTSA vPIC / CarQuery (VIN decode); Resend (email).

Static files: `express.static` for uploaded media under `uploads/` (inventory photos, etc.)